

**ONLINE PHARMACY MANAGEMENT SYSTEM**  
**(A CASE STUDY: SARANGA PHARMACY)**

**BY**  
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**A Research Proposal Submitted to The Department of Information Technology In Partial  
Fulfillment for the Requirement of Certificate in Information Technology Mount Kenya  
University**

**MAY 2023,**

## **DECLARATION**

### **Declaration by student**

I hereby declare that this project as my original work based on my creativity except in my references and acknowledgement section, and that the original work contain here has not been done by unspecified persons. This research proposal is my original work and has not been presented for a certificate in any other institution.

Signature.....

Date.....

Reg No: DIT/2023/49204

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### **Declaration by the Supervisor**

This research proposal has been submitted for examination with my approval as the university supervisor

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Date.....

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## **ABSTRACT**

Online pharmacy management system is a system of dispersing drugs on both small- and large-scale customers. The system is to be developed at Saranga dispensary. The pharmacy currently experiences the stock balance in which the pharmacist is not able to trace the number of drugs or if the required drug is available in the store, there is poor marketing of the drugs thus reduced selling. There is a problem with report generation at the pharmacy whereby the pharmacist is not able to keep track of the drugs sold thus leading to poor decision making when buying and selling of drugs. There are some of the solutions that may help in solving out the problem at Saranga pharmacy; the system will be able to alert the pharmacist the stock balance in the store, the system will be able to generate the good reports on time so as to improve the decision making and proper management of drugs. The main objective of the study is to design a system that manages drugs so as to improve on marketing and record keeping of available drugs. The modules to be used in the system are; registration module, login module, sales module and stock update module.

Data will be collected by interviewing some of the residents, use of questionnaires and by thorough investigation of the system study. The significance of this research is aimed at computerizing all the records concerning the drugs and the sales records. The system is used to make use of the rich internet technologies and consist of database which will capture, maintain and retrieve the drug records.

PHP and HTML languages will be used to design the Interface while MySQL will be used to design the database. The design tools that will be used will include Data Flow Diagrams and flowcharts. The graphical user interface is accessed by using passwords and the usernames are uniquely generated from the customers.

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## **LIST OF ABBREVIATION**

|             |                                   |
|-------------|-----------------------------------|
| <b>DFD</b>  | - Data Flow Diagram               |
| <b>EFT</b>  | - Electronic Fund Transfer        |
| <b>GUI</b>  | - Graphical User Interface        |
| <b>HTML</b> | - Hypertext Mark –up Language     |
| <b>PHP</b>  | - Hypertext Pre –Processor        |
| <b>SAP</b>  | - System Application and Products |

## **CHAPTER ONE**

### **INTRODUCTION**

#### **1.1. Background of the Study**

A system is the interrelated components that work together in order to archive a common goal. Management system is the process of planning, developing and managing by the use of the information system (Kemigham.). Pharmacy management system is a management system that is designed to improve accuracy and to enhance safety and efficiency in the pharmaceutical activities. It is a computer-based system which helps the Pharmacist to improve inventory management, cost, medical safety etc. The system allows the user to enter daily activities of a particular product or drug during opening stock. It also involves manual entry upon arrival of new batches of drugs and upon drug movement out of the pharmacy for a certain period.

The Sarange pharmacy was founded in the year 2009 to help patients in the interior region of Gucha district to have good access to drugs to enable them treat their health problem. The pharmacy has helped many people gain good health and also created job opportunities. The pharmacy membership is increasing daily and thus there is need for good record keeping and information flow of every day to ensure efficient running of the pharmacy.

Due to the size and quality service of the pharmacy, the pharmacy has a very large customer base. These customers tend to visit the pharmacy for services mostly when they close from work. At this period, the number of customers that patronize the pharmacy is on the increase, thereby making the workload of the pharmacists much more tedious. This case makes it difficult for the pharmacist to attend to customers in a short period. Meanwhile the pharmacist has to ensure satisfaction in services to keep their customers. The factors mentioned above, results in delay of the services being rendered to the customers, thereby slowing down sales and risk losing valuable customers in the long run.

## **1.2. Problem Statement**

Since the Saranga pharmacy uses the paper-based method of management, it has led to the following challenges in the pharmacy, delay in new drug registration thus the pharmacist is not able to know if a drug requested by a customer is available. This leads to more time wastage especially during peak hours for both the customer and the pharmacist who has to go and manually check for the drug in the store. Poor drug marketing-the pharmacy uses physical means in advertising their products whereby a customer has to come to the store to know the kind of drugs being sold. Out of stock medications as physical pharmacies often run out of stock of certain medications, which can be a problem for patients who rely on those medications.

### **1.3. Proposed Solution**

The proposed solution is to design a user-friendly system that eliminates the drawbacks of the existing system and alert the pharmacist the stock balance. Provide good record keeping, the system also will use usernames and passwords for authorized staff to enable security of data. A centralized database will be adapted to help eliminate cases of redundancy by allowing data to be used only when needed.

### **1.4. OBJECTIVE**

#### **1.2.1. Main objective**

The main aim of this project is to develop a pharmacy management system that will manage pharmacy activities so as to enhance good customer services and management of the pharmacy.

#### **1.2.2. Specific objectives**

- i. To design a system that will allow online registration of customers and reduce the delay in registration.
- ii. To design a system that will enable online drug enquiries in the pharmacy so as to improve customer services.
- iii. To design a system that will allow the customer to order drugs online.
- iv. To design a system that will enable generation of the reports so as to improve good decision making in the pharmacy.

#### **1.2.3. Research Questions**

- i. How will the system be able to allow customers to register online for improved services?
- ii. How will the system be developed to allow online drug enquiries?
- iii. How will the system be able to allow customers to order drugs online?
- iv. How will the system make good generation of reports to the pharmacy?

### 1.5. The Scope

This system will be designed for the Sarange pharmacy and can also be used by other pharmacy to enhance their management system. The system would have the following modules;

**The registration module;** which records all the personal details of the customers and enter them in the system.

**The login module;** which allows persons registered in the system to log in and to log out of the system.

**The sales module;** which allows the system users to purchase the drugs that they need.

**The stock update module;** which allows the pharmacist to add stock online for increased purchase.

### 1.6. Justification

The online pharmacy management system is a web-based system that will allow easy handling all pharmacy activities online easily and safely. Using interactive GUI anyone can easily learn to use the complete system. The system has improved on efficiency by automation of data entry and retrieval searching features that makes it easier to locate, view and update records regardless of the time it was used. There would be security issues for the system in terms of the user usernames and passwords. Furthermore, there is less time space occupied when they are filling their records since all the records will be filled in the system.

### **1.2.1. Project Risks and Mitigation**

Though digitalization is good, it has its own risks that are needed to be identified and addressed. Awareness of these risks can help to build mitigation measures into the project planning and management. The following are some of the risks and the mitigation: Regulatory compliance in the system as all regulations including those related to the sales prescription medications and protection of customers privacy. This will be curbed by establishing a strong compliance program that has a team of experts who are familiar with the relevant regulations and who are responsible for ensuring that the system complies with these regulations.

The system is web based and this means that it is subjected to unauthorized entry which can occur through hacking, computer viruses and ignorance among others. To curb this problem, the system will be secured using antiviruses, encryption of data, passwords and firewalls.

Operational risks in the system must be operated and maintained on a daily basis, they include human errors, equipment failures and natural disasters. This will be managed by implementing a disaster recovery plan that will outline how the system will be restored in the event of a disaster.

### **1.2.2. Significance of the study**

For the purpose of fully understanding the project without misinterpretation this report underlines and explains the significance of online pharmacy management system.

The system implementation values on the time spent in drug ordering for Saranga customers by the use of online pharmacy management system.

The system ensures that the staff of Saranga can track and view the registered and sold drugs for clients.

This project helps the researcher to gain system development and research skills. The project also helped in the contribution to future research in the same line.

The system mostly reduces the delay in the drug ordering process at Saranga where customers waste time travelling to purchase the drugs.

## CHAPTER TWO

### LITERATURE REVIEW

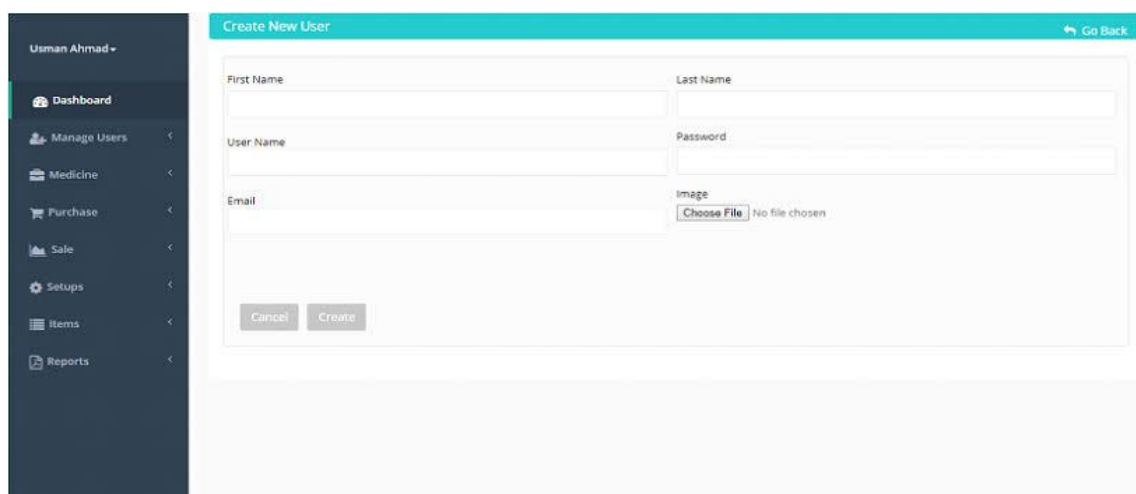
#### 1.7. Introduction

Literature review is the systematic identification, location and analysis of documents containing information related to the research problem being investigated. Its purpose is to increase confidence in your topic, information about the other researcher's works in your work area. It documents the state of the art with respect to the subject or topic you are writing about (Jokovic, 2019)

This chapter also describes the related information approaches of collecting information in using information system as a competitive advantage on other pharmacy services or systems. Good record keeping of drugs does not on rely on well trained IT support staff but also relies on the use of up-to-date developed system using the current ICT Technology. Without accurate up-to-date records of drugs, the pharmacy management may not be able to manage all their drugs data and information.

##### 1.2.1. Online Registration

According to (Orenstein, 2022) this module allows users to enter the required information including username, password, first name, last name, address, birth date and position. This will serve as the personal record of the users. The system will verify this information first before the authorized users can enter the system. Duplicating of users in the system will completely be eliminated due to a very rigid verification method.

The image shows a web application interface for creating a new user. On the left is a dark sidebar with a user profile 'Usman Ahmad' and a list of menu items: Dashboard, Manage Users, Medicine, Purchase, Sale, Setups, Items, and Reports. The main content area has a teal header 'Create New User' with a 'Go Back' link. Below the header is a form with several input fields: 'First Name', 'Last Name', 'User Name', 'Password', and 'Email'. There is also an 'Image' field with a 'Choose File' button and the text 'No file chosen'. At the bottom of the form are 'Cancel' and 'Create' buttons.

(Orenstein, 2022) Online registration module.

### 1.2.2. Online stock management system

According to (Shore, 2017) online management is the practice of ordering, storing, tracking and controlling inventory. Before industrial revolution when merchants had to write down purchases and keep an eye on how drugs were sold, and since they certainly didn't have a sale for casting, merchants have to forecast future needs themselves, which was not always accurate.

In early 1930s, a team of Harvard university designed what would become the first modern check out system. The university team came up with this type of stock management;



(Shore, 2017) online management stock issue list.

This system used punch cards which were inserted into a computer that read them and eventually passed the information to the store room in late 1960, the universal product code (UPC) was created, which improved stock management system, one of today is stock management is to the use of Radio microchips that transmit product information that contains everything that is relevant to business owner and the employees.

### **1.2.3. Design**

The design shall make use of both logical and physical design to show what happens from the time a customer purchases the drug and to a time the customer leaves the pharmacy. My SQL shall be used to develop the system database while PHP will be used to code the system. Guest's details could be accessed on the unique ID in the database thus enabling fast tracking of customer's details.

### **1.2.4. Coding**

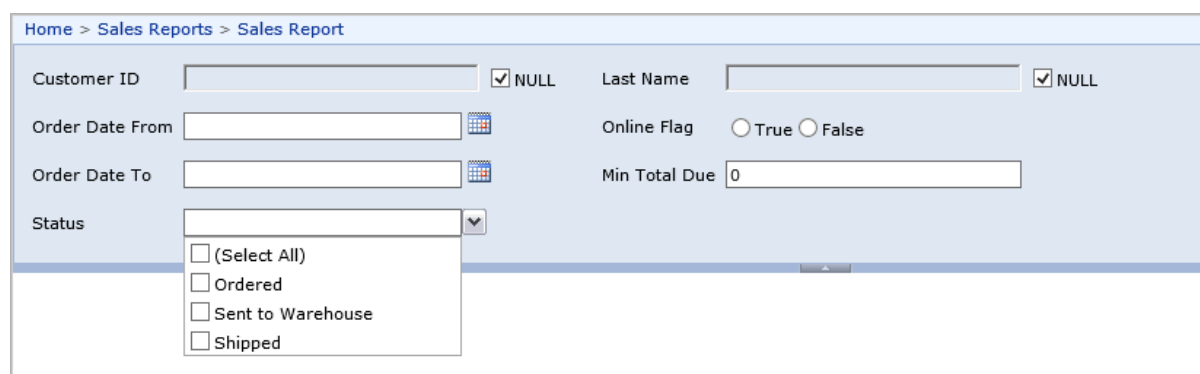
According to (Haillton, 2016) HTML which stands for Hypertext Markup Language is a predominant markup language for web pages, a building block of web pages. Web browsers read HTML documents and compose them into visual or audio web pages. The browser does not display the HTML tags but uses the tags to interpret the content of the page. Web browser can also refer to cascading style sheets (CSS) to define the appearance and layout of text and other materials (Taylor, 2019). PHP is a general-purpose scripting language originally designed for web development to produce dynamic web pages. For this purpose, PHP is embedded into the HTML source document and interpreted by a web server with a PHP processor module, which generates the web page document. SQL is a computer language designed for the management and retrieval of data in a database management system often referred to as Structured Query Language, designed for managing in RDMS.

### **1.2.5. Online display and purchase**

Online display is the displaying of products or services using digital technology such as the internet. According to (Patterson, 2022), online marketing is the process of leveraging web-based channels to spread messages about a product. Sarange pharmacy has been left behind from displaying, advertising or participating in social media or Google ad word for the purchase and marketing of their drugs. According to the study from (Sacha, Jeremy Hodara, 2012), Jumia has been able to display their different products and the respective amount hence receiving a wider market range. This has been so much beneficial since Jumia has been able to reach many people by the use of internet, emails, and other social media marketing thus being able to deliver their services to different people in the country.

### 1.2.6. Generation of report

Report-generation system is transparent to the metrology head used for data collection according to (MacAllister, Academic researchers, 2020). In the future, if the data acquisition system is changed, there will be practically no impact in the way the data are analyzed and reports are generated. A report generator is a computer program whose purpose is to take data from a source such as a database, or a spreadsheet, and use it to produce a document in a format which satisfies a particular human leadership. In pharmacy management system a report is an important document which needs to be generated to enable good decision making and management of the pharmacy.



(MacAllister, Free premade report template fit industry, 2020)

### 1.8. Research Gap

The Online pharmacy management System will develop to replace the manual process of all the activities carried out in the pharmacy. The old system does not serve the customer in a better way rather it makes customers' data vulnerable. The new system keeps proper records of customers for emergency and security purposes.

### 1.9. Conclusion

Finally, in online pharmacy management system, we have developed a user-friendly pharmacy management system. This system can take care of each member whether it's Owner or Customer. This system will help them to properly manage their pharmacy and help in growth without creating a hassle. This system is completely secure since every user is provided with user ID and password so there is no chance of unauthorized access. Online Purchasing, ordering and report generation make it easier to use. So, using this system will help in reducing the labor and providing more facility for customers to like the pharmacy and visit the pharmacy frequently.

## **CHAPTER THREE**

### **RESEARCH METHODOLOGY**

#### **1.10. Introduction**

Methodology refers to the analysis of the methods used appropriate to a field of study. It is a systematic way of accomplishing certain tasks and is defined as a collection of procedures, techniques, tools and documentation aids that helps system developer to speed up and simplify the system development process (Pressman, 2015). This will utilize multiple data collection methods such as observation, interviews and document reviews. To conduct this study, the researcher will collect data through questionnaires, observation and document review. The results will be later analysed and processed in a way that will help in the development of the system.

#### **1.2.1. Methods of Data Collection**

The following techniques are used to collect data; observation, questionnaires, interviews and document revision.

#### **1.3.1. Observation**

The team closely observe the pharmacy activities as the customers are served at the firm and how the management carried out its managerial activities. The pharmacy stock balance, alerting the expiry date of a drug, and generation of reports process is not well managed by the current system, hence, the team came up with this system to clearly outline the management boundaries of the pharmacy activities. The team also observe that the records were not well protected from unauthorised access to the other customer files by posing as the real customer.

#### **1.3.2. Questionnaires**

The researcher is sent to interview customers and the pharmacy firms staff members in to whom they would fill in the information that greatly to aid the team to come up with a system that would be user friendly to all the firm users. The questionnaires are conducted orally to the user who were present at the pharmacy firm during the time of the data collection and sent to the appropriate staff members who are away at the time of the data collection.

#### **1.3.3. Interview**

An interview is conducted between the researcher and the staffs of the pharmacy firm. The researcher created conducive environment in which the interviewees could feel free to let out the best they could help the team come up with the most effective system to fulfill their needs.

#### **1.3.4. Document Review**

Both the managers and the staff document are revised. Several data entry errors and retrieval problems were encountered. The system has put into consideration an error recovery strategy that will ensure maximum data integrity.

#### **1.3.5. Feasibility Study**

Preliminary investigations examine system feasibility; the likelihood the system will be useful to the organization. The main objective of the feasibility study is to test the Technical, Operational and Economical feasibility for adding new modules and debugging old running system.

#### **1.3.6. Technical Feasibility**

The feasibility will check whether the existing technical resources and manpower can help operate on the new system if not what extra resources are required so as to enable effective of the new online pharmacy management system. These resources are needed so as to meet the user requirements. The software includes windows operating system, html, php and a server which are needed to create website and store the needed information of participants. The pharmacy has got some existing computers which can be used by the administration.

#### **1.3.7. Operational Feasibility**

Operational feasibility was concerned whether the new system would be useful upon development and its implementation. It also checked whether the pharmacy administrators and the users were comfortable with use of the new system and whether it reduced the workload on the pharmacy since its computerized system and whether the system will solve the past problems

#### **1.3.8. Economic Feasibility**

Some of the importance for the development of the system is able to reduce the cost. Hence the system shall carry the economic study to determine whether the implemented system is cutting the expenditure the users spend or the system is increasing their daily expenditure. Also, the study shall be carried out to determine the cost of developing, operating, implementation and maintenance of the proposed system. If the system shall benefit both the users and the pharmacy firm then it should be implemented and used.

## 1.11. System Requirements

### 1.2.1. Hardware Requirements

Computer or laptop

Processor Speed: 2.0GHZ or any

RAM: 4 GB RAM or above

Hard Disk: 640 GB hard disk or above

HP laser jet printer will be required for printing purposes like invoice receipt.

USB flash disk (At least 2GB)

### 1.2.2. Software Requirements

**Operating System** at least latest will be required for efficient support of the system and its requirements.

**Microsoft Office 2016** will be required for typing purposes of the system for documentation

**SQL** will be required for database linking and programming development of the system.

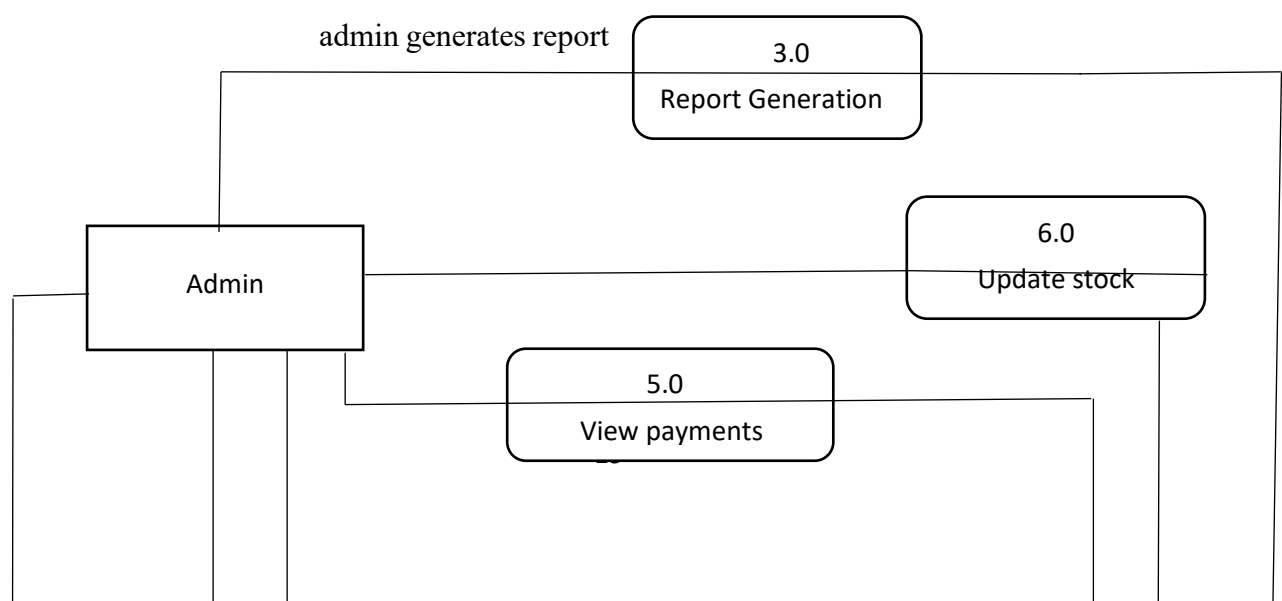
**Antivirus** will be required for preventing the system from being attacked by computer viruses during the development process.

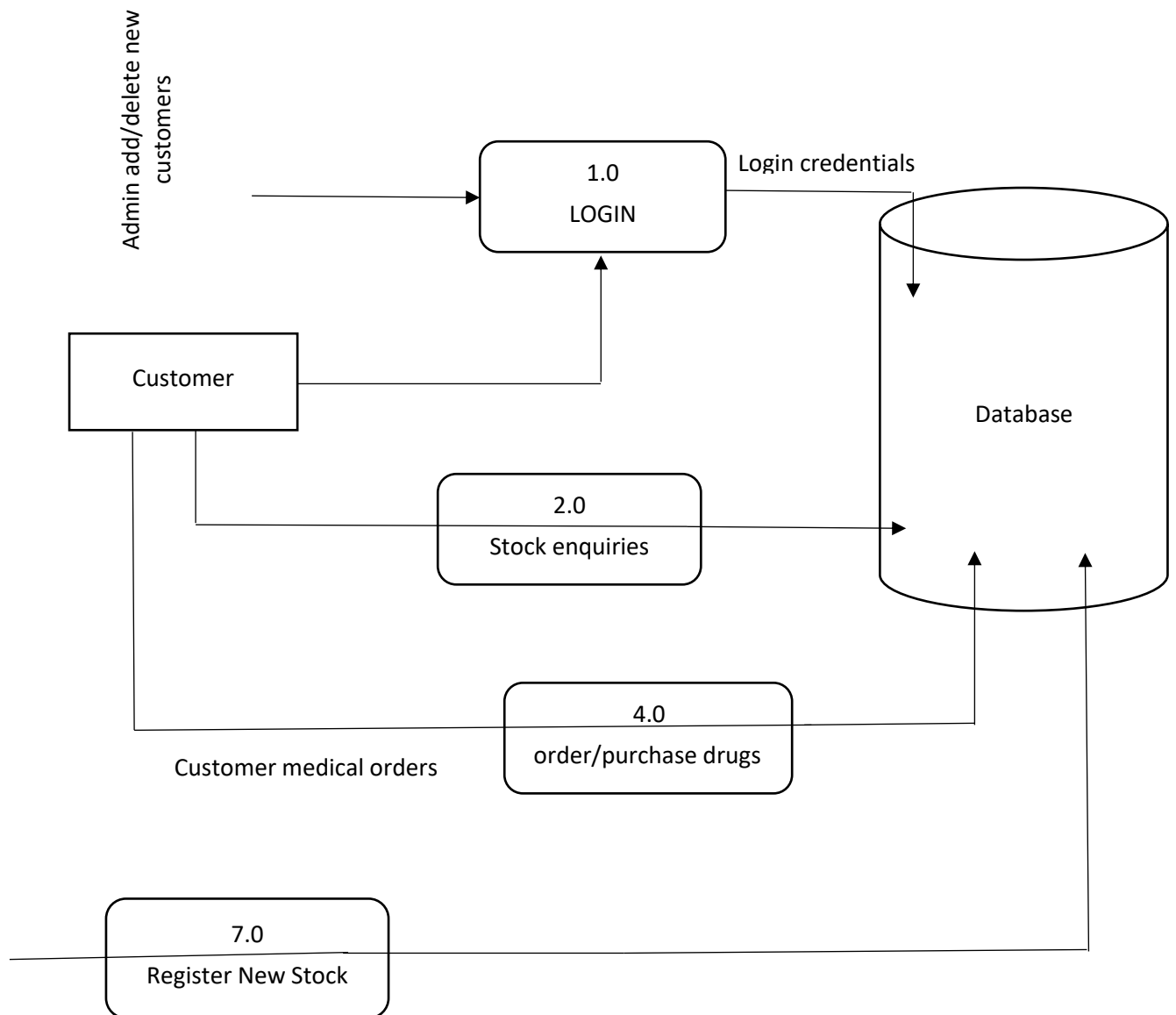
## 1.12. System Design

The proposed system will make use of both the logical and physical design. SQL will be used to develop the system database while PHP will be used to code the system.

## 1.13. System Design

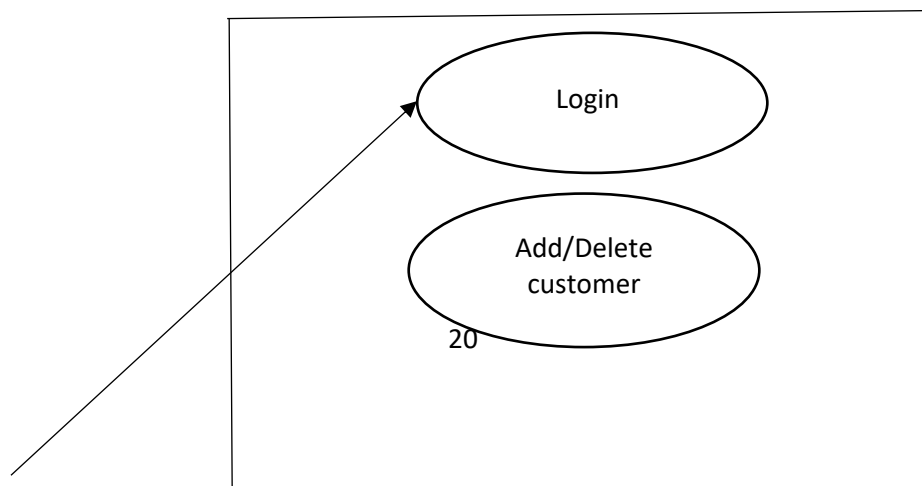
### 1.2.1. DFD for the proposed system

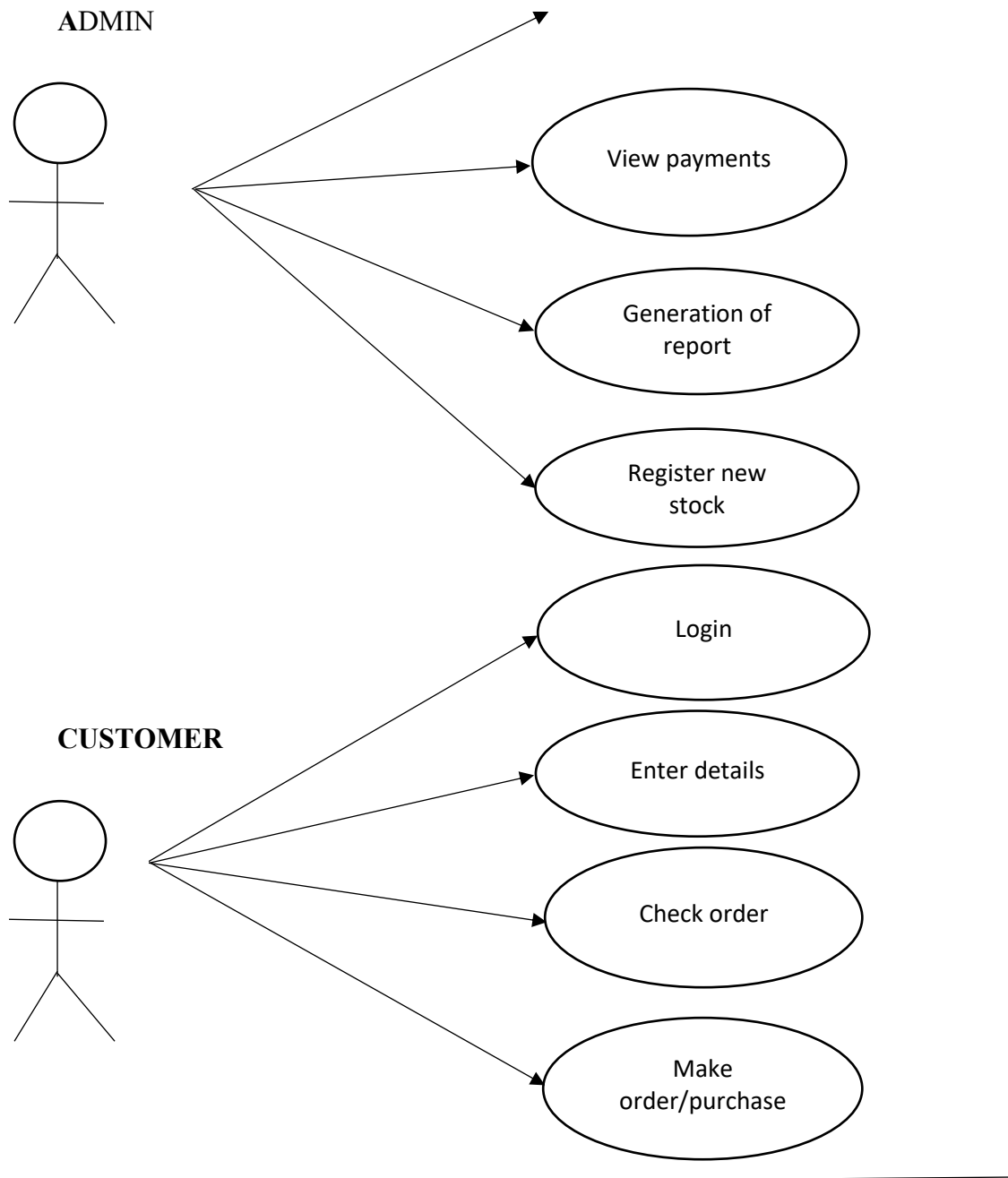




The dataflow diagram above is likely to show all the users' functionality, the admin is able to add the customer, update stock and report generation. The employee is able to make or enquire about their orders in the system easily.

### 1.2.2. User Case diagram





#### 1.14. System Testing

This is a level of system testing where a complete and integrated system tested. The purpose of this test is to evaluate the system's compliance with the specified requirements. Several strategies will be used in testing of the system. They include the following:

##### 1.2.1. Registration Module

**Functionality Testing:** Verify that users can register successfully with valid information and that appropriate error messages are displayed for invalid inputs.

**Usability Testing:** Assess the user-friendliness of the registration process and whether it is straightforward for users to complete the registration form.

**Security Testing:** Ensure that sensitive user data, such as passwords, is properly encrypted and stored securely to prevent unauthorized access.

**Performance Testing:** Test the registration process under various loads to check if it can handle multiple user registrations simultaneously without performance degradation.

### **1.2.2. Login Module**

**Functionality Testing:** Validate that users can log in using correct credentials and that appropriate error messages are shown for invalid login attempts.

**Security Testing:** Test for potential security vulnerabilities like SQL injection or session hijacking to ensure the login system is secure.

**Usability Testing:** Assess the user experience during the login process, including how clear and informative error messages are.

**Performance Testing:** Evaluate how the login system performs under different concurrent user loads to ensure responsiveness and stability.

### **1.2.3. Sales Module**

**Functionality Testing:** Check that all aspects of the sales process work correctly, including adding items to the cart, processing payments, and generating invoices.

**Integration Testing:** Verify the integration between the sales module and other modules like inventory and customer databases.

**Performance Testing:** Assess the system's response time during high sales periods to ensure it can handle peak loads.

**Security Testing:** Validate that financial transactions are conducted securely to protect customers' payment information.

### **1.2.4. Stock Update Module**

**Functionality Testing:** Confirm that the stock levels are accurately updated when sales occur or new stock is added to the inventory.

**Integration Testing:** Ensure the seamless integration of the stock update module with the sales module to maintain real-time stock information.

**Performance Testing:** Assess how the system handles a large number of stock updates simultaneously without delays or errors.

**Data Accuracy Testing:** Check that the stock quantities are correctly reflected in the system after updates.

#### **1.2.4. System Implementation**

This is concerned with the replacement of existing system with the new and from one way of doing things to another. There are three main methods but only one should be used; Direct changeover, parallel running and phase implementation.

System implementation was carried out by use of direct changeover that ensured smooth shift from paper-based mode publication to that of online based noticeboard.

#### **1.15. Limitations**

The following drawbacks were encountered when coming up with this research project.

Limited scope of testing: Testing will only be carried out in the environment it exists for example, testing in a Windows 10 Operating System only.

Some errors may be untraceable. Not all the required tests may be accomplished therefore the system will contain few problems in its operations. There will be bugs in start, which will need time to fix.

#### **1.2.1. Conclusion**

The online pharmacy system will help pharmacists who wish to use this system to manage their drug sales. The system will also store data that will be used for future reference in pharmacy management.

### 1.16. Budget

| NO: | ITEM                          | QUANTITY | COST PER UNIT | TOTAL COST |
|-----|-------------------------------|----------|---------------|------------|
| 1   | Computer, laptop              | 1        | 47500         | 47500      |
| 2   | Operating system (windows 10) | 1        | 5000          | 5000       |
| 3   | Anti-virus (SMADAV 2019)      | 1        | 1500          | 1500       |
| 4   | Eternal storage (CDs)         | 1        | 30            | 150        |
| 5   | USB flash disk (32gb)         | 1        | 2500          | 2500       |
| 6   | Hard disk (500gb)             | 1        | 7000          | 7000       |
| 7   | Printer                       | 1        | 15000         | 15000      |
|     | TOTAL                         |          |               | 78650      |

### 1.17. Gantt chart

|   | Activity Month                 | APRIL<br>2023 | MAY<br>2023 | JUNE<br>2023 | JULY<br>2023 | AUG<br>2023 | SEP<br>2023 | DEC<br>2023 |
|---|--------------------------------|---------------|-------------|--------------|--------------|-------------|-------------|-------------|
| 1 | Data collection and analysis   |               |             |              |              |             |             |             |
| 2 | Proposal writing/documentation |               |             |              |              |             |             |             |
| 3 | Design                         |               |             |              |              |             |             |             |
| 4 | Coding                         |               |             |              |              |             |             |             |
| 5 | Testing                        |               |             |              |              |             |             |             |
| 6 | Implementation                 |               |             |              |              |             |             |             |

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